Zürich
Switzerland

Nikola Milosavljević

Github Linkedin

Languages: Rust | Python | Go

nikolamilosa20@gmail.com,
(41) 78-348-32-24

Software Engineer specializing in distributed systems, scalability, and observability. Strong foundation in computer science and system design, with experience in cloud computing, designing autonomous AI agents and observability.

Experience

Site Reliability Engineer — **DFINITY Stiftung** — **Zürich** **11/2024 - Present**
Novi Sad **09/2022 - 11/2024**

Observability & automation - platform powering engineers with insights into thousands of machines at low cost and latency.

- Designed and implemented a scalable service discovery module powering the company's **observability platform**, synchronizing global system state, integrating with **Prometheus** and **VictoriaMetrics** and generating dynamic observability configurations.
- Architected the company's end-to-end logging infrastructure; led migration from AWS to on-prem Kubernetes, reducing costs by **95%** and supporting **30M logs/hour**.
- Constructed an automated release qualification pipeline, executing pre/post-upgrade performance tests to quantify regressions and guide release decisions.

Opencloud - frontier sovereign serverless cloud platform for AI agents. [opencloud.org]

- Engineered core components of the **control plane**, allowing on-demand cloud provisioning and insights.
- Piloted **AI nodes**, allowing customers to keep ownership of their data while providing powerful features for their **cloud engines**.
- Prototyped an **SRE agent**, giving engine owners helpful insights and automatic detections of anomalies happening on their platform.

Distributed Governance [dfinity/ic], [dfinity/dre]

- Automated node migrations that are reducing swapping time from **days** to **seconds** while improving security and flexibility.
- Introduced a linear optimization tool for **distributed network design**, enabling constraint-based topology generation and capacity planning for target network states.

Teaching assistant — **University of Novi Sad** — **Novi Sad** **09/2023 - 09/2024**

Microservice architecture

- Lead labs and assisted lectures for 300+ students across multiple university groups.
- Taught students concepts of gRPC, message queuing and saga patterns.

Introduction to cybersecurity

- Delivered introductory cybersecurity lectures to 150+ students, teaching core concepts.
- Held a hackaton, allowing students to learn and use penetration testing, basics of cryptography and public key infrastructure.

Fullstack Software Engineer (Outsorced) — **Schneider Electric** — **Novi Sad** **05/2021 - 09/2022**

- Built a billing determinant module integrating IoT devices and handling **terabytes of data** with reliable large-scale processing.
- Developed smart meter orchestration system enabling users to remotely configure, monitor and diagnose devices usiong well known protocols, improving observability and operation control at scale.
- Led a company wide infrastructure provisioning refactor, introducing automation with terraform and ansible.

Education

MSc in Computer Science — **Unvieristy of Novi Sad** — **09/2022 - 12/2023**

- Faculty of Technical Sciences
- GPA: **9.75/10** | **Cybersecurity and payment systems section**
- Graduated in the top of the class (in top **1%**).
- Thesis: *Recovering deleted files for forensics investigation*.

BSc in Computer Science — **Unvieristy of Novi Sad** — **09/2018 - 07/2022**

- Faculty of Technical Sciences
- GPA: **9.49/10** | **Computer software and automation section**
- Graduated in the top of the class (in top **1%**).
- Thesis: *breaking OWASP Juice Shop*.

Honors

National Fellowship — **University of Novi Sad, 2018–2022**

- Serbia's national fellowship for academic merit for top 1% of students, awarded for four consecutive years.

National CyberHero hackaton — **University of Belgrade, 2021**

- Placed **5th**, solving security challenges across multiple domains.

Stack

Databases: *SQL* (Postgres, MsSql), *NoSQL* (Neo4j, Mongo, Redis)
Cloud: AWS, GCP, Azure
Automation: GitHub Actions, Ansible, Bazel, Airflow
Containerization & Orchestration: Docker, Kubernetes